AC2204 Marking Scheme

Introduction to Management Accounting · Autumn 2022 · Paper 1

Examination Session and Year	Autumn 2022
Module Code	AC2204
Module Title	Introduction to Management Accounting
Paper Number	Paper 1
External Examiner	Professor Ivo De Loo
Head of the Department	Professor Mark Mulcahy
Internal Examiners	Dr Margaret Healy
Duration of Paper	1.5 hours
Special Requirements	Show your workings; marks will be awarded for workings. Separate answer books are not required.

Instructions to Candidates (as per exam paper): Answer three questions in total. Attempt Question One from Section A (compulsory) and TWO questions from Section B.

Own-figure rule: where a candidate makes an error at an early stage of a multi-step numerical question, full method marks are awarded for correctly applying the required technique to their own (incorrect) figure in all subsequent steps. Marks are never cascaded away for a single early error.

Marking philosophy: this is a closed-book, time-pressured examination. Academic citations are never required. Discursive answers are marked on depth of explanation and application to the specific scenario, not generic textbook recall.

Question Overview

Question	Scenario	Topics	Marks
Q1 (compulsory)	Tomy Huts Ltd	CVP analysis, breakeven, margin of safety, operating leverage, multi-product WACM expansion decision, CVP assumptions	40
Q2 (Section B option)	Plumtree Ltd	Marginal vs absorption costing with opening AND closing stock, profit reconciliation, ABC/leverage terminology	30
Q3 (Section B option)	RedTop Ltd	Two-stage activity-based costing (first-stage resource allocation), uses of ABC information, management vs financial accounting	30
Q4 (Section B option)	Statement evaluation	Prime vs conversion costs, differential vs sunk costs, high-low method, absorption vs marginal costing, cost structure vs sales mix	30

QUESTION 1

Tomy Huts Ltd

Total: 40 marks

Tomy Huts Ltd builds and installs wooden "Garden Office" structures. Candidates are required to (A) calculate and interpret five core CVP metrics, (B) evaluate a proposed second-product expansion ("Bed in the Shed") using the weighted average contribution margin approach, and (C) discuss two assumptions underlying CVP analysis.

Part	Requirement	Marks
(A)	Operating profit, breakeven (units and revenue), margin of safety, operating leverage – with explanation of usefulness	20
(B)	WACM breakeven for the two-product expansion; expected profit; advice on proceeding	14
(C)	Two CVP assumptions and their reasonableness	6

Question 1(A) **20 marks**

Tomy Huts Ltd: Variable cost per Garden Office unit €400; Sales price per unit €640; Annual fixed costs €12,000. Assume 60 units were sold in 2021. Calculate each of the following and explain their usefulness for financial planning: (i) Operating profit (loss); (ii) Breakeven level of sales in units; (iii) Breakeven level of revenues (using the contribution margin approach); (iv) Percentage margin of safety; (v) Operating leverage.

Full Model Answer

Contribution margin per unit = €640 − €400 = €240. Contribution margin ratio = €240 ÷ €640 = 37.5%.

(i) Operating profit – €2,400

Operating profit = (Units sold × contribution per unit) − fixed costs = (60 × €240) − €12,000 = €14,400 − €12,000 = **€2,400**. *Usefulness:* tells Sean the expected bottom-line profit at a realistic sales volume, forming the basis for budgeting and assessing whether the current single-product business is financially viable.

(ii) Breakeven level of sales in units – 50 units

Breakeven units = Fixed costs ÷ Contribution margin per unit = €12,000 ÷ €240 = **50 units**. *Usefulness:* shows the minimum number of Garden Offices that must be sold before the business breaks even, useful for setting minimum sales targets and assessing risk.

(iii) Breakeven level of revenues (contribution margin approach) – €32,000

Breakeven revenue = Fixed costs ÷ Contribution margin ratio = €12,000 ÷ 37.5% = **€32,000**. *Usefulness:* expresses the breakeven point in euro sales terms, useful for setting revenue targets and comparing directly against budgeted sales revenue.

(iv) Percentage margin of safety – 16.67%

Margin of safety % = (Budgeted sales − Breakeven sales) ÷ Budgeted sales = (60 − 50) ÷ 60 = 10 ÷ 60 = **16.67%**. *Usefulness:* indicates how far sales can fall from the budgeted 60 units before the company starts making a loss — a relatively low margin of safety here signals limited room for a sales shortfall and a higher-risk position.

(v) Operating leverage – 6.0

Operating (degree of) leverage = Total contribution margin ÷ Operating profit = (60 × €240) ÷ €2,400 = €14,400 ÷ €2,400 = **6.0**. *Usefulness:* indicates that a 1% increase in sales revenue would increase operating profit by approximately 6% — a high leverage figure showing that profit is very sensitive to changes in sales volume, consistent with the low margin of safety identified above.

Mark Allocation

Marks	Criteria
18–20	All five metrics correctly calculated, with a clear and specific explanation of the usefulness of each for financial planning (not generic statements).
13–17	Four or five metrics correctly calculated with reasonable explanations; one weaker or missing explanation.
8–12	Most metrics correctly calculated but explanations are generic or missing; or a systematic error affects several figures.
0–7	Limited understanding of CVP metrics; minimal correct workings.

Question 1(B) **14 marks**

Sean is considering launching "Bed in the Shed": selling price €800/unit, variable cost €500/unit. Extra production space rental of €5,160 per annum would be incurred. Sean expects to sell 2 Garden Office units for each Bed in the Shed unit sold. (i) Using the WACM approach, determine the breakeven point in units and sales revenues under the expansion proposal. (ii) If actual sales are 65 Garden Office units and 24 Bed in the Shed units, determine the expected operating profit (loss). (iii) Advise Sean on whether or not to proceed, giving one financial and one non-financial reason.

Full Model Answer – (i) WACM breakeven

Working – revised fixed costs and contribution per unit

Item	Value
Existing fixed costs	€12,000
Extra production space rental	€5,160
Total fixed costs under the expansion proposal	**€17,160**
Garden Office contribution margin per unit (€640 − €400)	€240
Bed in the Shed contribution margin per unit (€800 − €500)	€300

Working – weighted average contribution margin (sales mix 2 Garden Office : 1 Bed in the Shed)

Item	Value
Contribution per 'basket' of 3 units (2 × €240 + 1 × €300)	€780
WACM per unit (€780 ÷ 3)	**€260.00**

Breakeven point – units and revenue

Item	Value
Total breakeven units (€17,160 ÷ €260)	66 units
Garden Office breakeven (66 × 2/3)	44 units
Bed in the Shed breakeven (66 × 1/3)	22 units
Breakeven revenue (44 × €640 + 22 × €800)	**€45,760**

Alternative revenue approach: Breakeven revenue can equally be found via the weighted average contribution margin ratio: basket revenue of €2,080 (2 × €640 + 1 × €800) gives a WACM ratio of €780 ÷ €2,080 = 37.5% (coincidentally the same ratio as the single-product case in part (A)); €17,160 ÷ 37.5% = €45,760, the same result.

Full Model Answer – (ii) Expected operating profit at actual sales

The 2:1 sales mix ratio used in part (i) is only an assumption for the WACM breakeven calculation. Actual expected unit sales (65 Garden Offices and 24 Bed in the Shed units) are given directly, so profit is calculated straightforwardly from those actual figures.

Working – expected operating profit under the expansion

Item	€
Garden Office contribution (65 units × €240)	15,600
Bed in the Shed contribution (24 units × €300)	7,200
Total contribution margin	**22,800**
Less: Total fixed costs under the expansion proposal	(17,160)
Expected operating profit (with expansion)	**5,640**

Comparison note: This compares to an operating profit of only €2,400 without the expansion (from part (A)(i)) — the expansion increases expected profit by €3,240, more than doubling it, even though the actual sales mix (65:24, approximately 2.7:1) differs somewhat from the 2:1 mix assumed for the breakeven calculation in part (i).

Full Model Answer – (iii) Advice on the expansion proposal

Financial reason

Proceeding with the expansion is expected to increase operating profit from €2,400 to €5,640, an increase of €3,240 (more than double the current profit level) at the stated sales volumes. On this basis, the expansion appears financially attractive, provided the sales estimates are reasonably reliable.

Non-financial reason

The initial market research indicating strong demand for Bed in the Shed provides some non-financial support for proceeding, reducing (though not eliminating) uncertainty around the sales estimates used above. Against this, Sean should also weigh non-financial risks such as the operational complexity of managing a second product, reliance on the newly rented production space, and the inherent uncertainty of market research forecasts, which may not be borne out in practice.

Overall advice: On balance, given the significant projected profit improvement and supportive market research, Sean should be advised to proceed with the expansion, while continuing to monitor actual demand closely against the assumptions used.

Mark Allocation

Marks	Criteria
13–14	WACM breakeven correctly calculated in units and revenue; expected profit at actual sales correctly calculated (recognising the actual mix differs from the WACM assumption); clear advice with one distinct, well-reasoned financial and one non-financial factor.
9–12	Substantially correct workings for (i) and (ii) with a minor error; reasonable advice given in (iii).
5–8	WACM approach understood but applied with errors; advice generic or one-sided without balancing considerations.
0–4	Limited understanding of the multi-product breakeven approach; minimal correct workings or advice.

Question 1(C) **6 marks**

Identify and discuss two assumptions underlying CVP analysis and consider the extent to which they can be held as reasonable approximations of reality.

Full Model Answer

1. Costs can be reliably classified as fixed or variable, and behave in a linear fashion within the relevant range

CVP analysis assumes selling price, variable cost per unit, and total fixed costs all remain constant across the range of activity being considered, so total cost and total revenue can be represented as straight lines. In reality this is only an approximation: variable costs per unit may fall due to bulk purchase discounts on materials or rise due to overtime premiums at high volumes, and fixed costs can 'step up' once a certain level of activity is exceeded (e.g. needing to rent further production space, as in part (B)). The assumption is reasonable within a limited relevant range of activity over a short time horizon, but becomes less reliable for large changes in volume.

2. The sales mix remains constant (for multi-product businesses)

Where more than one product is sold, breakeven analysis using the WACM approach (as applied in part (B)) assumes products are sold in a constant, predictable ratio to one another. In reality, sales mix can and does shift due to changing customer preferences or seasonality, which changes the weighted average contribution margin and therefore the breakeven point. This is illustrated directly in part (B), where the actual expected sales mix (65 Garden Offices to 24 Bed in the Shed units, approximately 2.7:1) differs from the 2:1 mix assumed for the breakeven calculation — showing this assumption is often only an approximation of actual trading conditions.

Other acceptable assumptions: Candidates may alternatively discuss: all units produced are sold (no inventory build-up); or that the analysis holds only within a defined relevant range of activity outside of which cost/revenue relationships are assumed to change. Any two well-explained assumptions, discussed with reference to their reasonableness, receive full credit.

Mark Allocation

Marks	Criteria
5–6	Two clearly identified assumptions, each explained accurately and with a genuine, specific discussion of how reasonable the assumption is in practice.
3–4	Two assumptions identified with reasonable explanation; limited discussion of their reasonableness.
1–2	One assumption identified and explained, or two assumptions stated with minimal explanation.
0	No relevant assumptions identified.

QUESTION 1 TOTAL — 40 MARKS

QUESTION 2

Plumtree Ltd

Total: 30 marks

Plumtree Ltd manufactures and sells a single product and, unlike a first-year company, carries 400 units of opening stock into the year, valued differently under marginal and absorption costing. Candidates are required to (A) prepare income statements under both costing methods, (B) reconcile the resulting profit figures, and (C) explain two management accounting terms.

Part	Requirement	Marks
(A)	Unit product cost, closing stock value and operating profit – marginal and absorption costing	17
(B)	Profit reconciliation note	7
(C)	Product-cost cross-subsidization and degree of operating leverage	6

Question 2(A) **17 marks**

Plumtree Ltd, year to 30 December 2021: Sales 3,000 units; Selling price €26.00; Production 3,200 units; Variable manufacturing costs (total) €42,240; Fixed manufacturing overheads (total) €19,360; Variable selling expenses (total) €10,000; Fixed selling expenses (total) €6,000. 400 units in opening stock at 1 Jan 2021, valued at €5,024 (marginal costing) and €7,120 (absorption costing). (A) Calculate the unit product cost, the value of closing stock and the operating profit using (i) Marginal costing and (ii) Absorption costing.

Full Model Answer

Preliminary workings

Closing stock = Opening stock + Production − Sales = 400 + 3,200 − 3,000 = 600 units. Variable manufacturing cost per unit (current year) = €42,240 ÷ 3,200 = €13.20. Fixed manufacturing overhead per unit (current year, absorption costing only) = €19,360 ÷ 3,200 = €6.05. The opening stock values (€5,024 marginal, €7,120 absorption) are given and reflect the prior period's cost rates — they are not recalculated using this year's unit costs. Closing stock, however, is valued at this year's unit cost, since it consists of units produced this year.

(i) Marginal costing

Unit product cost and closing stock – marginal costing

Item	€
Unit product cost (variable manufacturing cost only, current year)	13.20
Closing stock value (600 units × €13.20)	**7,920**

Income statement – marginal costing

Item	€
Sales revenue (3,000 units × €26.00)	78,000
Opening stock (given)	5,024
Add: Variable cost of production (3,200 units × €13.20)	42,240
Less: Closing stock (600 units × €13.20)	(7,920)
Variable cost of goods sold	39,344
Less: Variable selling expenses	(10,000)
Contribution margin	**28,656**
Less: Fixed manufacturing overheads	(19,360)
Less: Fixed selling expenses	(6,000)
Operating profit (marginal costing)	**3,296**

Full Model Answer – (ii) Absorption costing

Unit product cost and closing stock – absorption costing

Item	€
Unit product cost (€13.20 variable + €6.05 fixed, current year)	19.25
Closing stock value (600 units × €19.25)	**11,550**

Income statement – absorption costing

Item	€
Sales revenue (3,000 units × €26.00)	78,000
Opening stock (given)	7,120
Add: Full production cost (3,200 units × €19.25)	61,600
Less: Closing stock (600 units × €19.25)	(11,550)
Cost of goods sold	57,170
Gross profit	**20,830**
Less: Variable selling expenses	(10,000)
Less: Fixed selling expenses	(6,000)
Operating profit (absorption costing)	**4,830**

Mark Allocation

Marks	Criteria
15–17	Both unit product costs, both closing stock values, and both fully-formatted income statements correct, correctly incorporating the given opening stock values alongside current-year production costs.
11–14	Substantially correct with a minor computational error in one figure, or one method fully correct and the other largely correct.
6–10	Correct approach to at least one costing method; opening stock handled incorrectly (e.g. recalculated using current-year rates) or other errors in workings/presentation.
0–5	Limited understanding of the distinction between the two costing methods; minimal correct workings.

Question 2(B)

7 marks

Prepare a short note reconciling the profit figures from Part (A) above and explaining why this difference occurs.

Full Model Answer

Why the profit figures differ

Absorption costing carries fixed manufacturing overhead forward in inventory (as part of the unit product cost), whereas marginal costing expenses fixed manufacturing overhead in full as a period cost. The profit difference between the two methods in any period is therefore explained entirely by the net movement of fixed overhead **into and out of** stock during the period: fixed overhead carried forward in this year's closing stock increases absorption costing profit relative to marginal costing profit, while fixed overhead that was carried in opening stock (from the prior period) and is now released against this period's sales reduces absorption costing profit relative to marginal costing profit for the current period.

Profit reconciliation

Item	€
Operating profit – marginal costing	3,296
Add: Fixed manufacturing overhead carried forward in closing stock (600 units × €6.05)	3,630
Less: Fixed manufacturing overhead released from opening stock (€7,120 − €5,024)	(2,096)
Operating profit – absorption costing	**4,830**

Note on the opening stock fixed overhead figure: The €2,096 fixed overhead component of opening stock is found as the difference between the absorption and marginal costing values given for opening stock (€7,120 − €5,024), since this represents the fixed overhead per unit that was absorbed into those units in the *prior* period (at the prior period's own fixed overhead rate, which is not the same as this year's €6.05 rate).

Mark Allocation

Marks	Criteria
6–7	Clear explanation of the overhead deferral/release mechanism and a correct, clearly presented profit reconciliation using both the closing and opening stock fixed overhead movements.
4–5	Good explanation with a substantially correct reconciliation; a minor error, e.g. omitting or mis-signing the opening stock adjustment.
2–3	Some understanding shown of why the figures differ, but the reconciliation is materially incomplete or incorrect.
0–1	No meaningful explanation or reconciliation provided.

Question 2(C) **6 marks**

Explain each of the following terms used in management accounting: (i) Product-cost cross-subsidization; (ii) Degree of operating leverage.

Full Model Answer

(i) Product-cost cross-subsidization

Product-cost cross-subsidization occurs when a costing system's overhead allocation method causes some products to be overcosted and others undercosted relative to the overhead they actually cause to be incurred, so that the overcosted products effectively subsidise the undercosted ones. It typically arises under traditional, volume-based absorption costing when a single driver (such as machine hours or labour hours) is used to allocate overhead that is not actually driven by production volume, distorting reported unit costs and potentially leading to poor pricing or product-mix decisions.

(ii) Degree of operating leverage

The degree of operating leverage (DOL) measures how sensitive operating profit is to a given percentage change in sales revenue or volume. It is calculated as total contribution margin divided by operating profit at a given level of activity. A higher DOL indicates that a business's cost structure is weighted more heavily towards fixed costs relative to variable costs, meaning profit is more sensitive (and riskier) to changes in sales volume — a small percentage change in sales produces a proportionately larger percentage change in operating profit.

Mark Allocation

Marks	Criteria
5–6	Both terms accurately and fully explained, correctly distinguishing the definition and practical implication of each.
3–4	Both terms explained with reasonable accuracy; one explanation less developed.
1–2	One term explained well; the other term explained only partially or generically.
0	No meaningful explanation of either term.

QUESTION 2 TOTAL — 30 MARKS

QUESTION 3

RedTop Ltd

Total: 30 marks

RedTop Ltd operates a chain of restaurants and has initiated a two-stage activity-based costing study. Candidates are required to (A) perform the first-stage allocation of resource costs to activity cost pools and compute activity rates, (B) state two potential uses of this information, and (C) discuss whether management and financial accounting are meaningfully different despite drawing on the same underlying data.

Part	Requirement	Marks
(A)	First-stage cost allocation and activity rates for each cost pool	15
(B)	Two potential uses of the activity rate information	5
(C)	Discuss whether shared data means no real difference between management and financial accounting	10

Question 3(A) 15 marks

RedTop Ltd, month of July. Activity cost pools and levels: Meal preparation (2,600 meals); Ingredient preparation, including storage (1,000 cubic feet); Customer service (15 restaurants); Other (not allocated). Resource costs and their distribution across the four pools: Food ingredients €48,000 (80%/15%/5%/0%); Kitchen supplies €3,000 (39%/35%/15%/11%); Chefs' salaries €20,000 (65%/15%/12%/8%); Equipment depreciation €7,000 (5%/35%/50%/10%); Restaurant staff wages €8,000 (18%/12%/60%/10%); Administration costs €15,000 (0%/10%/28%/62%). (A) Perform the first stage allocation of costs and compute activity rates for each of the cost pools.

Full Model Answer

The first stage of this two-stage ABC system allocates each resource cost (the rows) across the four activity cost pools (the columns) according to the percentage distribution given, based on resource consumption.

Working – first-stage allocation of resource costs to activity cost pools (€)

Resource cost	Meal Prep.	Ingredient Prep.	Customer Service	Other
Food ingredients (€48,000)	38,400	7,200	2,400	0
Kitchen supplies (€3,000)	1,170	1,050	450	330
Chefs' salaries (€20,000)	13,000	3,000	2,400	1,600
Equipment depreciation (€7,000)	350	2,450	3,500	700
Restaurant staff wages (€8,000)	1,440	960	4,800	800
Administration costs (€15,000)	0	1,500	4,200	9,300
Total allocated to each pool	**54,360**	**16,160**	**17,750**	**12,730**

Check: Total allocated across all four pools = €54,360 + €16,160 + €17,750 + €12,730 = €101,000, agreeing with the sum of all six resource costs given (€48,000 + €3,000 + €20,000 + €7,000 + €8,000 + €15,000 = €101,000).

Working – activity rates

Cost pool	Pool total (€)	Activity level	Activity rate
Meal preparation	54,360	2,600 meals	€20.91 per meal
Ingredient preparation	16,160	1,000 cubic feet	€16.16 per cubic foot
Customer service	17,750	15 restaurants	€1,183.33 per restaurant
Other	12,730	Not allocated	No activity rate calculated

Note on the 'Other' pool: The €12,730 allocated to the 'Other' cost pool represents business-sustaining costs that cannot be meaningfully traced to meals, storage space, or individual restaurants using a sensible cost driver, and is therefore deliberately left unallocated in a well-designed ABC system, rather than being forced onto an arbitrary driver.

Mark Allocation

Marks	Criteria
13–15	All six resource costs correctly allocated across all four pools; pool totals correctly summed and checked against the total resource cost; all three calculable activity rates correctly computed; 'Other' pool correctly left unallocated.
9–12	Substantially correct first-stage allocation with minor computational errors in a small number of cells or rates.
5–8	General approach to first-stage allocation understood but with multiple errors, or activity rates not correctly computed from the pool totals.
0–4	Limited understanding of the two-stage ABC allocation process; minimal correct workings.

Question 3(B) **5 marks**

State and explain two potential uses of the information in your answer to Part (A) above.

Full Model Answer

> **1. Benchmarking restaurants and identifying best practice**
> Because the activity rates are calculated on a company-wide basis, RedTop Ltd can compare each individual restaurant's actual resource consumption (meals prepared, storage space used, customer service demands) against these average rates to identify restaurants that are operating more or less efficiently than the norm. This directly supports the stated management objective of learning from best practices at individual restaurants, allowing efficient locations to be studied and their practices shared with underperforming locations.

> **2. More accurate cost information for pricing and menu decisions**
> The activity-based rates provide a more accurate picture of what actually drives cost (meal volume, storage space, and restaurant-level customer service demand) than a simple, single overhead rate would. This can better inform decisions such as menu pricing, evaluating the profitability of individual restaurants, or deciding whether to open, close, or resize particular locations, since managers can see which cost pools are the largest drivers of total cost.

Mark Allocation

Marks	Criteria
4–5	Two distinct, well-explained uses, each clearly connected to the specific information generated in part (A) (activity rates and cost pool breakdown).
2–3	Two uses identified with reasonable explanation, or one use explained in excellent depth.
1	One use identified with limited explanation.
0	No relevant uses identified.

Question 3(C)

10 marks

Discuss the following statement, outlining why you agree or disagree: "As management accounting and financial accounting use the same basic data set of organisational information, there is really no difference between the two."

DISAGREE

While management accounting and financial accounting both ultimately draw on the same underlying transactional data recorded within an organisation, this does not mean there is no meaningful difference between the two disciplines. The difference lies not in the raw data itself, but in how that data is selected, processed, regulated, and used.

Users and purpose

Financial accounting processes the shared data into standardised reports for external users (shareholders, lenders, tax authorities) who need a reliable overall picture of performance and position. Management accounting selectively draws on and supplements the same underlying data — and often generates entirely new data not captured in the financial ledgers at all, such as the activity cost driver volumes (meals, cubic feet, restaurant counts) used in part (A) — to produce internal reports tailored to specific management decisions.

Regulation and format

Financial accounting is bound by external regulatory requirements (company law, IFRS/GAAP) governing exactly how the shared data must be presented. Management accounting is entirely unregulated and can present the same underlying data in whatever format and at whatever level of detail (e.g. per-restaurant, per-meal, per-cubic-foot, as in this question) is most useful for the decision at hand — a level of granularity financial accounting would never require or permit in its standardised statements.

Time orientation and scope

Financial accounting is fundamentally historical. Management accounting, while it also uses historical data (as in this ABC study), places significant additional emphasis on forward-looking analysis such as budgeting, forecasting, and decision support, extending well beyond what the shared historical data set alone can provide.

Conclusion: The statement conflates the *source* of the data with the *discipline* that processes it. A shared starting data set does not eliminate the substantial differences in purpose, regulation, format, and time orientation that distinguish management accounting from financial accounting — the statement should therefore be rejected.

Mark Allocation

Marks	Criteria
8–10	Clear disagreement, well argued, covering at least three distinct points of difference (users/purpose, regulation/format, time orientation/scope) with specific reference to how RedTop Ltd's data could be used differently by each discipline.
5–7	Correct disagreement with two well-explained points of difference.
2–4	Some relevant discussion but generic, lacking depth, or only one point of difference clearly made.
0–1	Agrees with the statement, or no meaningful discussion provided.

QUESTION 3 TOTAL — 30 MARKS

QUESTION 4

Statement Evaluation

Total: 30 marks

Five independent statements covering prime vs conversion costs, differential vs sunk costs, the high-low method, absorption vs marginal costing, and cost structure vs sales mix. For each statement, candidates must state whether they agree or disagree, with supporting calculations and explanation.

Part	Topic	Marks
(A)	Prime costs vs conversion costs	6
(B)	Differential costs vs sunk costs	6
(C)	Bravo Ltd – high-low method cost estimate	6
(D)	Absorption costing product cost definition	6
(E)	Cost structure vs sales mix	6

Question 4(A)

6 marks

Prime cost consists of all direct manufacturing costs and is the same thing as conversion costs.

Full Model Answer

DISAGREE

The first part of the statement is broadly correct — prime cost is the total of direct manufacturing costs (direct materials, direct labour, and any direct expenses). However, prime cost is **not** the same thing as conversion cost.

> ### Prime cost
> Prime cost = Direct materials + Direct labour (+ any direct expenses). It represents all costs directly and exclusively traceable to the product being manufactured.

> ### Conversion cost
> Conversion cost = Direct labour + Manufacturing overhead. It represents the cost of converting raw materials into a finished product.

> ### Why they are not the same
> Prime cost and conversion cost share direct labour in common, but they differ in their other component: prime cost includes direct materials and excludes manufacturing overhead, while conversion cost includes manufacturing overhead and excludes direct materials. A cost (direct materials) can therefore be part of prime cost without being part of conversion cost, and a cost (manufacturing overhead) can be part of conversion cost without being part of prime cost — the two measures are distinct, not interchangeable.

Mark Allocation

Marks	Criteria
5–6	Correct disagreement with accurate definitions of both prime cost and conversion cost, correctly identifying direct labour as the only common element and explaining why the two totals otherwise differ.
3–4	Correct disagreement with accurate definitions of both terms; the shared/differing components not explicitly explained.
1–2	Correct disagreement but with a vague or partially inaccurate definition of one term.
0	Incorrect verdict or no meaningful explanation.

Question 4(B) **6 marks**

Differential costs are costs that have already been incurred and must always be considered when choosing one alternative over another.

Full Model Answer

DISAGREE

The description confuses differential cost with sunk cost. Differential costs are **not** costs that have already been incurred — that describes a sunk cost. The claim that differential costs 'must always be considered' is correct in principle, but the underlying definition given is wrong.

> **What a differential cost actually is**
> A differential cost (also called an incremental cost) is the **difference** in total cost between two alternative courses of action. Differential costs are future-oriented and relevant to decision-making precisely because they change depending on which alternative is chosen; they have not yet been incurred at the time the decision is being made.

> **Why 'already incurred' describes a sunk cost instead**
> A cost that has already been incurred and cannot be changed by any future decision is, by definition, a sunk cost. Sunk costs are the same regardless of which alternative is chosen going forward, which is exactly why they are irrelevant to decision-making — the opposite of a differential cost, which is relevant precisely because it does differ between alternatives.

Mark Allocation

Marks	Criteria
5–6	Correct disagreement, correctly identifying that the statement's definition actually describes a sunk cost, with accurate definitions of both differential cost and sunk cost given.
3–4	Correct disagreement with an accurate definition of differential cost; the sunk cost mix-up not explicitly identified.
1–2	Correct disagreement but with a vague or partially inaccurate explanation.
0	Incorrect verdict (agrees with the statement) or no meaningful explanation.

Question 4(C)

6 marks

Monthly activity levels and associated costs at Bravo Ltd: HIGH: October, 210 units, total cost €14,631. LOW: April, 70 units, total cost €11,397. If it is planned to produce 180 units in January next year, then based on the cost function for the most recent financial year, the expected cost is €12,451.

Full Model Answer

DISAGREE

The correct expected cost at 180 units is €13,938, not the €12,451 stated.

Working – high-low method

Item	Value
Variable cost per unit ((€14,631 − €11,397) ÷ (210 − 70))	(€3,234 ÷ 140) = €23.10 per unit
Fixed cost (at high point: €14,631 − (210 × €23.10))	€9,780
Fixed cost (at low point: €11,397 − (70 × €23.10)) – check	€9,780
Cost function	Total cost = €9,780 + (€23.10 × units)

Estimated cost at 180 units (January)

Item	Value
€9,780 + (€23.10 × 180)	€9,780 + €4,158
Correct estimated cost	**€13,938**

Note: The €12,451 figure in the statement does not correspond to any obvious alternative calculation from this data and appears to simply be an incorrect figure; candidates are not required to identify the source of the error, only to demonstrate the correct cost function and resulting estimate.

Mark Allocation

Marks	Criteria
5–6	Correct disagreement; variable cost per unit and fixed cost correctly calculated and cross-checked at both points; correct cost function applied to give €13,938.
3–4	Correct disagreement with a minor computational error, or the cross-check at the second point omitted.
1–2	High-low method attempted with the correct general approach but a significant computational error.
0	Incorrect verdict (agrees with the statement) or no meaningful workings.

Question 4(D)

6 marks

Absorption costing treats only those costs of production that vary with output as product costs.

Full Model Answer

DISAGREE

This description is the definition of **marginal (variable) costing**, not absorption costing.

> **What absorption costing actually treats as product cost**
>
> Absorption costing treats **all** costs of production as product costs — both variable manufacturing costs (which do vary with output) **and** fixed manufacturing overhead (which does not vary with output). Fixed manufacturing overhead is absorbed into each unit produced using a predetermined absorption rate and carried in inventory until the unit is sold, exactly as demonstrated for Plumtree Ltd in Question 2, where the €6.05 fixed manufacturing overhead per unit is included in the €19.25 absorption unit cost.

> **What the statement is actually describing**
>
> Treating only those production costs that vary with output as product cost, and excluding fixed manufacturing overhead entirely (charging it instead as a period cost), is exactly what **marginal costing** does. The statement has the two methods' defining characteristics reversed.

Mark Allocation

Marks	Criteria
5–6	Correct disagreement with an accurate description of what absorption costing actually includes (both variable and fixed production costs), and a correct identification that the statement describes marginal costing instead.
3–4	Correct disagreement with a reasonable description of absorption costing; the marginal costing mix-up not explicitly identified.
1–2	Correct disagreement but with a vague or partially inaccurate explanation.
0	Incorrect verdict (agrees with the statement) or no meaningful explanation.

Question 4(E)

6 marks

Cost structure refers to the relative proportions in which an organisation's products are sold.

Full Model Answer

DISAGREE

The description given in the statement is the definition of **sales mix**, not cost structure.

What cost structure actually is

Cost structure refers to the relative proportion of fixed costs and variable costs within an organisation's total costs. A cost structure weighted heavily towards fixed costs (capital intensive) produces higher operating leverage and a lower margin of safety than one weighted towards variable costs (labour intensive), for the same total revenue and total costs.

What the statement is actually describing

The relative proportions in which an organisation's different products are sold is the definition of **sales mix**. Sales mix is central to multi-product breakeven analysis using the weighted average contribution margin approach, as applied to Tomy Huts Ltd's Garden Office and Bed in the Shed expansion in Question 1. Cost structure and sales mix are both important CVP concepts, but they describe entirely different things: one relates to the fixed/variable split of costs, the other to the mix of products sold.

Mark Allocation

Marks	Criteria
5–6	Correct disagreement with an accurate definition of cost structure and a correct identification that the statement's description actually matches sales mix.
3–4	Correct disagreement with a reasonable definition of cost structure; the sales mix point not explicitly identified.
1–2	Correct disagreement but with an inaccurate or vague definition of cost structure.
0	Incorrect verdict (agrees with the statement) or no meaningful explanation.

QUESTION 4 TOTAL — 30 MARKS